Merlin Keerthana K A

Data Analyst

Chennai, Tamil Nadu | +91 79046 72450 | merlink470@gmail.com
Portfolio: https://merlin-data-analyst.github.io/portfolio/ LinkedIn: http://www.linkedin.com/in/merlin-keerthana
GitHub : https://github.com/merlin-data-analyst

PROFESSIONAL SUMMARY

Analysed **3,630+** QSR Quick Service Restaurant operational records to identify trends, uncover patterns and generate actionable business insights as a Data Analyst. Developed interactive Power BI dashboards with KPI cards, navigation buttons and performance metrics for QSR operations and e-commerce sales. Applied machine learning techniques for supply chain logistics predictive analysis and conducted an A/B testing case study using statistical hypothesis testing. Utilised SQL, Python, Excel and Power BI for data cleaning, exploratory data analysis (EDA), trend analysis, data visualisation and performance reporting to support data-driven decision-making.

TECHNICAL SKILLS

Programming & Libraries: Python, Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Scikit-learn, Statsmodels, ETL (Pipeline).
Database & SQL: MySQL, Joins and Subqueries, Window Functions, Aggregate Functions, CTEs.
BI & Visualization: Power BI, Tableau, DAX, Power Query, Data Modelling, Dashboard Development, KPI Reporting, PowerPoint.
Spreadsheets: Microsoft Excel/Google Sheets: Pivot Tables, XLOOKUP, Data Cleaning, Spreadsheet formulas, Charts, Data Validation.
Core Competencies: Data Analysis, Data Cleaning, Root Cause, Data Visualization, Statistical Analysis, Hypothesis Testing, A/B Testing, Correlation Analysis, Trend Analysis, Predictive Analysis, Business Intelligence, Data Storytelling, Reporting, Data-Driven Decision-Making, Data Warehouse.

PROJECTS

Supply Chain Logistics Analysis using Machine Learning | Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn | Link : View-Project

- Analysed supply chain logistics data and uncovered a delivery delay rate of around **82.1%**, identifying delays as the central operational problem.
- Cleaned and transformed raw data with Pandas and NumPy, then ran EDA and correlation analysis showing minimal geographic differences and several variables with near-zero correlation to key outcomes.
- Trained and evaluated Scikit-learn models and found the available features had limited predictive power for some targets.
- Translated analytical findings into a data storytelling narrative, documenting key insights, data limitations and actionable recommendations to support logistics performance monitoring and future data-driven decision-making.

QSR Operational Analysis | Python, MySQL, Power BI, Excel | Link : View-Project

- Analysed Quick Service Restaurant (QSR) operational data to evaluate waiting times, order duration, staff efficiency, customer satisfaction and inventory availability.
- Cleaned and transformed **3,630** records using Python, identifying long waiting times in **15.2%** of orders and stock unavailability in **25%** of orders.
- Evaluated operational relationships, identifying a negative correlation of **- 0.306** between waiting time and customer satisfaction and a positive correlation of **0.578** between staff efficiency and customer satisfaction.
- Developed an interactive Power BI dashboard with multiple analytical pages, navigation buttons and a question-based exploration feature to monitor operational KPIs and support data-driven decision-making.
- Recommended optimising staff allocation during peak hours, improving inventory replenishment practices and monitoring waiting-time KPIs to enhance operational efficiency and customer satisfaction.

E-commerce Data Analytics | Python(Pandas), MySQL, Power BI, Google Sheets | Link : View-Project

- Transformed e-commerce data using Python Pandas and analysed sales performance, customer behaviour and product trends using SQL queries in MySQL.
- Evaluated key business metrics, including total and average sales, **top 10** products by revenue, best-selling products by quantity, regional sales, return quantity, total orders and total customers.
- Recommended prioritizing top-performing products, strengthening sales strategies in high-performing regions such as the United Kingdom and investigating product returns to identify opportunities to improve revenue, reduce returns and support data-driven business decisions.

Data Analyst | Case Study | Link : View-Project

- Conducted an A/B testing case study comparing black and white website backgrounds, identifying a **14.07%** higher observed conversion rate for the black-background variant white around **5.40%** and recommending data-driven website optimization based on conversion performance.

PROFESSIONAL EXPERIENCE

Trade Fair Associate | ACMEE (AIEMA), Ambattur, Chennai **May 2025 – Jun 2026**

- Maintained and organised exhibitor datasets, improving data accuracy and consistency through data cleaning and validation, while supporting efficient data management across exhibitor records.
- Resolved exhibitor queries and operational issues for **500+ exhibitors**, demonstrating strong stakeholder reporting and management, effective communication and problem-solving skills while ensuring smooth coordination and timely issue resolution.
- Supported international machine tool exhibition operations and coordinated exhibitor information.

EDUCATION

University of Madras (Correspondence) | Master of Computer Applications (MCA) **2024 – 2026 (Currently Pursuing)**

Chevalier T Thomas Elizabeth College for Women | Bachelor of Computer Applications (BCA) **2022 – 2024**

T.J.S Polytechnic College | Diploma in Computer Science Engineering (DCSE) **2019 – 2022**

CERTIFICATIONS

Google Data Analytics Professional Certificate | Google, Coursera (Online)

- Certificate: https://coursera.org/share/47275382af45a78ff5138b29449ab1f6